

HYBRID HUMOR DETECTION IN CODE-MIXED TENGLISH

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submitted by

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DECLARATION

I undersigned hereby declare that the project report **Hybrid Humor Detection in Code-Mixed Tenglish** submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology in the Computer Science & Engineering, SRM University-AP, is a bonafide work done by me under supervision of Dr. V R P S SASTRY YADAVILLI. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree of any other University.

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ABSTRACT

Humor detection in code-mixed languages is difficult because of language differences, inconsistent transliterations, and informal expressions often used in digital communication. This work focuses on detecting humor in Romanized Telugu-English (Tenglish) text, which is common on social media platforms. Traditional Natural Language Processing (NLP) models often have trouble with this type of data due to its lack of grammatical structure and mixed-language patterns. To tackle these challenges, we propose a hybrid method that combines transformer-based models with a rule-based heuristic engine. We used a balanced dataset of 10,000 samples, including both humorous and non-humorous text. This dataset was split into training, validation, and testing sets using an 80:10:10 division. We evaluated model generalization with two dataset versions (V2 and V3) through cross-dataset experiments. We tested multiple transformer models, including MuRIL, mBERT, XLM-RoBERTa, IndicBERT, IndicBERT v2, Telugu-BERT, XLM, LaBSE, mT5, and ByT5. Among these, MuRIL showed the best performance because it had been pretrained on Indian languages and transliterated text. The addition of the heuristic engine further improved results by capturing non-linguistic cues such as slang, sarcastic patterns, and style variations.

CONTENTS

ACKNOWLEDGMENT	i
ABSTRACT	ii
LIST OF TABLES	v
LIST OF FIGURES	vi
Chapter 1. INTRODUCTION TO THE PROJECT	1
1.1 Background of the Study	1
1.2 Introduction to Humor Detection	2
1.3 Code-Mixed Language and Tenglish	2
1.4 Challenges in Tenglish Humor Detection	3
1.5 Role of Transformer Models in NLP	3
1.6 MuRIL and Its Importance	4
1.7 Need for a Hybrid Approach	4
1.8 Problem Statement	4
1.9 Objectives of the Project	5
1.10 Scope of the Project	5
1.11 Organization of the Report	5
Chapter 2. MOTIVATION	6
Chapter 3. LITERATURE SURVEY	9
3.1 Introduction to Literature Survey	9
3.2 Traditional Machine Learning Approaches	9
3.3 Deep Learning-Based Approaches	10
3.4 Transformer-Based Models	11

3.5	Indian Language Models and MuRIL	11
3.6	Code-Mixed Language Processing	12
3.7	Humor Detection in Social Media	12
3.8	Hybrid Approaches	13
3.9	Cross-Dataset Evaluation	13
3.10	Research Gaps	14
3.11	Summary of Literature Survey	14
Chapter 4.	DESIGN AND METHODOLOGY	16
4.1	Overview of the Proposed System	16
4.2	System Architecture	16
4.3	Data Preprocessing	17
4.4	Transformer-Based Model Design	18
4.5	Heuristic Engine Design	18
4.6	Hybrid Prediction Mechanism	19
4.7	Algorithm Workflow	20
4.8	Handling Code-Mixed Text	20
4.9	Design Considerations	20
4.10	Summary of Design and Methodology	21
Chapter 5.	IMPLEMENTATION	22
5.1	Overview of Implementation	22
5.2	Development Environment	23
5.3	Dataset Preparation	23
5.4	Text Preprocessing	24
5.5	Transformer Model Implementation	24
5.6	Heuristic Engine Implementation	25
5.7	Hybrid Integration	26
5.8	Training Process	26
5.9	Evaluation Metrics Implementation	27
5.10	Cross-Dataset Evaluation Implementation	27

5.11 System Deployment	27
5.12 Summary of Implementation	28
Chapter 6. HARDWARE AND SOFTWARE TOOLS USED	29
6.1 Introduction	29
6.2 Hardware Requirements	29
6.3 Software Requirements	30
6.4 Development Tools	31
6.5 Libraries and Frameworks	31
6.6 Operating System	32
6.7 Summary of Tools Used	32
Chapter 7. RESULTS & DISCUSSION	33
7.1 Introduction	33
7.2 Transformer Model Comparison	33
7.3 Graphical Analysis	34
7.4 Cross-Dataset Evaluation	35
7.5 Confusion Matrix Analysis	36
7.6 Discussion of Results	37
7.7 Summary	38
Chapter 8. CONCLUSION	39
REFERENCES	40

LIST OF TABLES

7.1	Performance Comparison of Transformer Models	34
7.2	Cross-Dataset Evaluation Results	36
7.3	Confusion Matrix (MuRIL + Heuristic)	37

LIST OF FIGURES

4.1	Proposed Hybrid Architecture for Humor Detection	17
7.1	Accuracy comparison across transformer models	35
7.2	F1 score comparison across transformer models	35
7.3	Precision vs Recall comparison	36
7.4	Cross-dataset evaluation results	36

Chapter 1

INTRODUCTION TO THE PROJECT

1.1 BACKGROUND OF THE STUDY

As digital communication technologies rapidly evolve, the way people communicate and share information with each other is changing dramatically. Social media sites, messaging apps, and online bulletin boards have quickly become the primary forms of communication for many individuals, particularly younger generations. In multilingual nations like India, this change has led to many instances of code-mixing, in which speakers of two or more languages mix languages into one sentence or conversation. One of the most common types of code-mixed language is Tenglish (Telugu + English written in Roman script).

The use of code-mixing can be seen as an extension of the natural patterns of communication among humans, where speakers of two or more languages will switch from one language to another depending on the context, convenience and expressiveness. In addition, code-mixing creates numerous challenges for Natural Language Processing (NLP) systems. Most existing NLP systems have been developed with the monolingual and grammatically correct nature of standard written texts in mind; therefore, NLP systems typically do not perform well when parsing informal, ungrammatical, or linguistically diverse data like Tenglish.

Detecting humor in text represents one of the most complex challenges faced by both humans and NLP systems because it often requires analysing

(or interpreting) the text for more than just its literal meaning but also subtle contextual cues such as sarcasm, irony, hyperbole (exaggeration), and references to culture. When the challenge of understanding humor in code-mixed language is combined by the additional challenges of spelling (i.e. using different letter combinations to represent the same word in different languages) and informal grammar, and the fact that expressions can vary widely between different cultures, this presents a very significant challenge for NLP systems.

1.2 INTRODUCTION TO HUMOR DETECTION

Detection of humor is one of the areas of interest in the study of NLP because of its use in sentiment analysis, chatbots, content moderation, and social media analytics. Automatic detection of humor is a difficult task due to the subjective nature of humor. Humor usually requires implicit knowledge, cultural sensitivity, and linguistic skill. In textual communication, there are various ways through which humor may be communicated; these include wordplay, sarcasm, irony, exaggeration, and situational humor. All of these require some sort of contextual knowledge which cannot be achieved through rule-based models. Machine and deep learning algorithms have shown promising results in this regard.

1.3 CODE-MIXED LANGUAGE AND TENGLISH

Code-mixing is a linguistic phenomenon where words, phrases, and even grammatical elements from two different languages are mixed within a sentence. India has many examples of code-mixing, especially since there are various languages spoken along with English in India. Tenglish is one

such example of a code-mixed language that is a combination of Telugu and English. There are some special characteristics associated with Tenglish text that need to be considered when designing NLP algorithms. The first characteristic relates to phonetically spelled words where the same word can have different spellings based on the pronunciations of speakers.

1.4 CHALLENGES IN TENGLISH HUMOR DETECTION

The identification of humor in Tenglish text entails numerous difficulties. First, the absence of spelling standardization renders it tough to identify the words properly. Second, the use of multiple languages in a single sentence makes syntactic and semantic analyses challenging. Third, humor is usually culturally dependent and contextualized, hence making it hard for computational algorithms to detect such aspects. Apart from the above-mentioned difficulties, Tenglish text usually contains emojis, slang terms, elongated words, and repetition of letters. These elements help to convey humor, but most of the NLP tools have neglected them.

1.5 ROLE OF TRANSFORMER MODELS IN NLP

However, transformer architectures have changed the face of NLP by providing better means for handling contextual information. Architectures like BERT, XLM-RoBERTa, and mBERT make use of self-attentive networks to represent relationships between words in a sentence; thus, they provide better means for handling context compared to older methods. In the case of multilingual transformer architectures, they provide the capacity of handling several languages using one model architecture. However, most of these architectures are trained on formal texts and hence may lack

effectiveness when handling informal, code-mixed texts.

1.6 MURIL AND ITS IMPORTANCE

MuRIL stands for Multilingual Representations for Indian Languages. This is a transformer-based architecture designed particularly for Indian languages along with their transliterations. In contrast to general multilingual models, MuRIL has been trained on datasets that consist of code-mixed and transliterated texts. This renders it more capable of handling Tenglish. Given its capabilities in dealing with phonetic variations and mixed languages, MuRIL becomes an effective choice for Tenglish humor detection. Nevertheless, MuRIL might still fail to identify non-linguistic humor, such as emojis, which requires other techniques.

1.7 NEED FOR A HYBRID APPROACH

To address the shortcomings of pure data-driven approaches, this project adopts a combination of transformer models and rule-based heuristic engine. The transformer model takes care of context and semantic features, whereas the heuristic engine aims at detecting humor-related cues like emojis, slang, sarcasm, and styles. The use of both these modules helps to capitalize on deep learning techniques as well as domain-specific expertise.

1.8 PROBLEM STATEMENT

Though tremendous progress has been made in natural language processing, effective solutions that can detect humor in mixed code languages like Tenglish have not yet been found. These models are unable to account

for various transliterations, slangs, and cultural contexts. Hence, a more reliable approach is required to detect humor in such linguistic environments.

1.9 OBJECTIVES OF THE PROJECT

The key aim of this research is the design of a hybrid humor detection system that can detect humor in Tenglish text through transformers and heuristic approaches. This system is required to be highly accurate and robust.

1.10 SCOPE OF THE PROJECT

This project covers the following activities: preparing datasets, training of the models, evaluation of several transformers, and creating an intelligent engine. Cross-dataset evaluation will be conducted for assessment purposes.

1.11 ORGANIZATION OF THE REPORT

This report consists of several chapters. The first chapter contains an introduction to the research and its background. Other chapters include literature review, data set description, method, experiment, results, discussion, limitation, future work, and conclusion.

Chapter 2

MOTIVATION

The rise in digital communication has changed the way people communicate their thoughts, feelings, and ideas. Social media platforms such as WhatsApp, Instagram, and Twitter have become major modes of communication, especially amongst the youth generation. Multilingual countries such as India have adopted various forms of code-switching in communication, which involves combining more than one language in a sentence. Such examples include Tenglish, a blend of the Telugu language and English language in Roman script. Communication through such code-mixed languages is expressive and dynamic and makes NLP research very interesting.

Humor is vital in online communication because it engages users, builds social ties, and communicates emotions well. Meme sharing, jokes, sarcasm, and funny posts are the most shared content in any online platform. Nevertheless, humor identification is still a tough task to do automatically. Humor requires implicit meanings, cultural knowledge, and linguistic skills that make it hard for computer algorithms to recognize it.

The problem becomes even more difficult when code-mixed languages like Tenglish come into play. Unlike regular written text, Tenglish does not follow any fixed grammatical syntax and relies on phonetically spelled texts that are different for each individual. The same Telugu word could be expressed in different ways depending upon its pronunciation and how it is typed. In addition to this, there is always a use of slang, abbreviations,

viations, emojis, elongated words, and repetitions in these kinds of texts. All these factors play an important role in providing humor but are usually ignored by NLP models.

All existing humor detection models are only applicable for datasets written in one language. Although these models are quite successful, their application cannot be extended to code-mixed languages because of the reasons mentioned above. Models like multilingual transformer models XLM-RoBERTa, BERT, and mBERT have helped a lot in handling multiple languages, but they cannot cope up with transliterations and informalities.

The development of transformers such as MuRIL, which is trained using Indian languages and their transliterations, presents an encouraging path to resolving these issues. MuRIL can handle phonetic differences and understand mixed languages better than general-purpose multilingual models. Nevertheless, sophisticated models such as MuRIL might fail to capture non-linguistic signals, such as emojis, sarcasm, and stylistic cues, necessary for humor recognition.

Such gaps provide grounds for adopting a hybrid model combining transformers and heuristic engines. Transformers will analyze the semantic and contextual content of the Tenglish text, while heuristic engines will extract further information regarding the text's humorous elements. Such integration between the two parts will enable the system to have a complete grasp of humor in Tenglish text.

The third reason that motivates us to perform this research is the unavailability of public data sets for humor detection in code-mixed languages. All available data sets lack large numbers of entries or have a mono-lingual format. We address this problem through developing a dataset of Tenglish samples with an equal proportion of humorous and non-humorous data.

Such a dataset facilitates the process of training and evaluating the model effectively.

Additionally, the practical importance of humor detection cannot be underestimated since humor can be applied in numerous areas outside scientific research. Humor detection can be useful for sentiment analysis, developing chatbots, moderating content, or recommendation systems. For example, identifying a humorous input would allow conversational agents to respond accordingly. At the same time, a similar technique can be employed by content moderators to filter out jokes from other types of content.

Moreover, cross-dataset testing is another aspect of evaluating the robustness of NLP models because real-world data usually contains variability in terms of language usage and contexts, which should be taken into consideration to ensure proper generalization. In this respect, this study utilizes multiple versions of datasets for evaluating the proposed model through cross-dataset testing. This way, it would be possible to make sure that the model performs well when applied to different scenarios.

On balance, the main idea behind this research is grounded in the necessity to overcome the drawbacks of current NLP models in regard to detecting humorous text in code-mixed languages, especially Tenglish. Applying advanced models of transformers and heuristics to this task allows overcoming current difficulties related to such types of texts and developing an effective model of humorous text recognition.

Chapter 3

LITERATURE SURVEY

3.1 INTRODUCTION TO LITERATURE SURVEY

Humor detection has become one of the most important research questions in NLP due to its various applications in sentiment analysis, chat-bots, social media monitoring, and human-computer interaction. The difficulty of automatic humor detection is that it requires contextual knowledge, cultural references, and connotations. Various solutions to this problem have been suggested, such as machine learning approaches and more recently proposed deep learning approaches. The rapid rise in multilingual or code-mixed languages has presented a new aspect of the problem. Many countries like India have numerous languages, and people usually combine multiple languages within one sentence to form new language, for instance, Telugu and English make Tenglish. This is a new challenge to the humor detection problem, as the traditional approaches are unable to solve it. This chapter focuses on the previous work on humor detection, code-mixed languages, and transformers.

3.2 TRADITIONAL MACHINE LEARNING APPROACHES

In the past, scientists have primarily used traditional machine learning approaches (like Support Vector Machines (SVM), Naïve Bayes classifiers, and Decision Trees) for humor classification. The primary input to these

systems was a set of features that were hand-crafted by an individual or group (such as word frequency counts or n-grams/text patterns, part of speech tagging, sentiment scoring). Although each individual feature could provide a small piece of information regarding whether or not a given text (sample) was humorous, they fell short when trying to capture more sophisticated contextual meaning. Many forms of humor contain complex linguistic patterns (such as sarcasm and irony) that are difficult to capture with simple feature engineering techniques. As a result, traditional machine learning methods have been limited in their ability to identify humor and often yield poor results when evaluating complex and informal data sets.

3.3 DEEP LEARNING-BASED APPROACHES

Deep learning has led to increased usage of neural network models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) for Natural Language Processing (NLP). CNNs are good for identifying local features and patterns in text while RNNs (like Long Short-Term Memory (LSTM) networks) are good at modelling sequential dependencies.

As a result of using automated methods of feature representation learning from data, these models have improved detection of humour. However, when working with these models, there are limitations in terms of long-range dependency and context relationships. Large amounts of labelled data are necessary for proper model training, though there may not be large amounts of data available for code-mixed languages.

3.4 TRANSFORMER-BASED MODELS

Innovative developments in natural language processing (NLP) have been brought about by the advent of transformer architectures, most notably exemplified by BERT (Bidirectional Encoder Representations from Transformers), which employ self-attention mechanisms to enable the model to better understand the relationships between words in context. Consequently, transformers have provided previously unavailable levels of contextual benefit to NLP models relative to other types of technology. In addition, extensions of these models such as XLM-RoBERTa and mBERT, allow multiple languages to be processed by a single framework for multilingual analysis. These advances have been validated across multiple forms of NLP tasks including sentiment analysis, question-answering, and text classification.

However, while transformer-based architectures have achieved significant advancements to date, most were trained primarily on structured/formal sources of text and therefore are often not as effective at processing informal/code-mixed data such as Tenglish (hybridization of Tamil and English). This has increased the necessity for models/techniques designed specifically to address these limitations.

3.5 INDIAN LANGUAGE MODELS AND MURIL

Multilingual models are not ideal when it comes to working with Indian languages so specialized models like MuRIL were created. MuRIL is trained on a large diverse dataset of Indian languages, as well as transliterated text, which makes it an excellent option for working with code-mixing input. With MuRIL being able to work with phonetics as well as multi-

ple languages, it has a significant advantage over non-specialized models. MuRIL has proven to perform better than other multilingual models on various types of tasks where specifically Indian languages are involved. However, like all transformer models, MuRIL may have challenges with capturing non-linguistic cues (for example, emojis) and/ or stylistic features.

3.6 CODE-MIXED LANGUAGE PROCESSING

With the ubiquity of code-mixing in social media communication, many researchers have been working on developing methods to process it. Examples of this include techniques such as language identification, transliteration normalisation and feature extraction. Due to the absence of standardised data sets, there are many challenges in building code-mixing natural language processing (NLP) systems. Variability in spelling and grammar makes generalising across different examples difficult for model development. Using annotated data sets and creating specialised preprocessing techniques have been two areas addressed by different studies.

3.7 HUMOR DETECTION IN SOCIAL MEDIA

When considering a broad array of humorous material, such as memes, jokes, and sarcastic remarks, there are abundant online sources available through social media platforms from which researchers can derive large datasets for studying how humans detect humour. Previous studies have predominantly been based on the English language and therefore may not be relevant for non-English speakers as well as in multilingual contexts.

Social media text also presents a number of challenges because it is typically informal, which includes the use of slang, abbreviations and emojis

to express humorous content. These informal characteristics have a major impact on conveying humor, but traditional models of humour detection do not typically account for these types of variants.

3.8 HYBRID APPROACHES

There has been investigations done on hybrid models which combine machine learning / deep learning type models with traditional rule-based systems to create a composite that takes advantage of the benefits of both styles of model building by bringing together domain knowledge from both data-driven learning as well as model-based learning.

With respect to detecting whether something is humorous, heuristic rules can be employed to establish specific patterns in humour such as emojis, repeated characters, and indicators of sarcasm, ect. By utilizing these same heuristic rules along with transformer models, the combination of the two will enhance the overall performance of humour detection in addition to capturing additional sources of information that would be more difficult to derive using a data-driven approach alone.

3.9 CROSS-DATASET EVALUATION

Cross-dataset evaluation is a critical component of natural language processing research, as this type of evaluation measures how well a model will generalize beyond the training dataset. Since so many studies in the field have been conducted using only a single dataset to evaluate model performance, these results may not accurately represent how well a model would perform in the real world.

In particular, when working with code-mixed bilingual or multilin-

gual processing applications, cross-dataset evaluation plays a crucial role because of the wide range of variability in how code-mixing occurs in real life. In order for models to be able to generalize to real-world scenarios, they must be able to account for these differences in distribution within datasets.

3.10 RESEARCH GAPS

Although NLP has advanced considerably in recent years, there are still many gaps when it comes to detecting humor in code-mixed languages. The majority of studies to date have used only monolingual data, failing to capture the complexities of handling both languages simultaneously. Furthermore, little work has been done on integrating transformer models and heuristic techniques to improve model performance when detecting humor in code-mixed languages. Another important gap is that evaluations of all models across multiple datasets have been limited; this means that researchers have not evaluated how well their models perform across different types of data, making it difficult to assess their ability to generalize well.

3.11 SUMMARY OF LITERATURE SURVEY

The literature survey highlights the evolution of humor detection techniques from traditional machine learning methods to advanced transformer-based models. While significant progress has been made, challenges remain in handling code-mixed languages such as Tenglish. The need for specialized models, robust datasets, and hybrid approaches is evident. This project aims to address these challenges by combining transformer models with heuristic techniques and evaluating their performance across multiple datasets. The proposed approach builds upon existing re- search

while introducing novel contributions to the field of code-mixed humor detection.

Chapter 4

DESIGN AND METHODOLOGY

4.1 OVERVIEW OF THE PROPOSED SYSTEM

The intended system is designed for detecting humour in both Romanized Telugu-English Code-Mixed Text using a hybrid approach that incorporates a combination of transformer based models and a rule-based heuristic engine. The problem addressed by this system is related to the informal/casual nature of the language in Code Mixed text, including transliteration inconsistencies, the use of slang and stylistic varieties.

Initially, input text will be processed and then contextual features will be extracted from the input using transformer models. At the same time, a heuristic engine will be used the same text input features, to gather additional signals indicative of humour. Both outputs from each component will be combined to produce the final classification result. The overall result of the hybrid design of the system will provide both capabilities of deep learning and domain knowledge.

4.2 SYSTEM ARCHITECTURE

There are three primary elements that make up the architectural layout of the anticipated system: (1) processing of input data; (2) applying transformer technology to create a classification method; and (3) conducting an analysis by applying heuristic methods to analyze input data for

humor. A depiction of the system architecture can be seen in Figure 4.1.

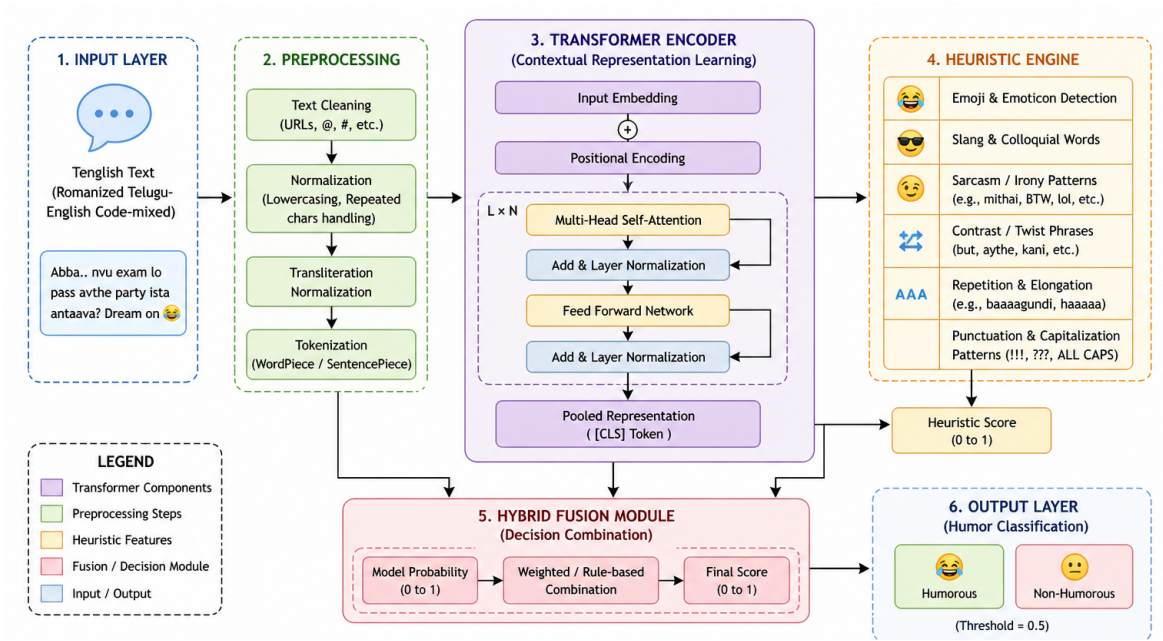


Figure 4.1: Proposed Hybrid Architecture for Humor Detection

After removal of added noise, pre-processing and formatting will take place before tokenization of all input text data will occur. The tokenized data will then be processed through a transformer model to create contextual embeddings from each token while simultaneously processing input data through a heuristic engine to look for patterns in the data attributed to humour. Both transformed and heuristic outputs will be combined to create a final classification output.

4.3 DATA PREPROCESSING

The effectiveness of a model can be greatly influenced by the pre-processing of data that are either non-formal or a mixture of different languages (code-mixed). The normalisation of text, removal of redundant characters, and resolution of special character issues can all contribute to this.

Tenglish text has many different transliteration variations, so normalising the variations can decrease inconsistencies and improve the speed and effectiveness of the models as well. All emojis and special characters are kept when the pre-processing adds value in terms of humour recognition; they should not be discarded. Each transformer model has its own dedicated tokenizer and that tokenizer is used to perform tokenisation before applying any of the pre-trained embedding models.

4.4 TRANSFORMER-BASED MODEL DESIGN

For feature extraction and classification, transformer models are a primary use-case. The self-attention mechanism in the transformer models allows them to capture the contextual relationships between the words in a way that helps deepen our understanding of texts.

Several transformer models are examined in this project: MuRIL, mBERT, XLM-RoBERTa, IndicBERT, IndicBERT v2, Telugu-BERT, XLM, LaBSE, mT5, and ByT5. All of the models were fine-tuned for binary classification (to classify texts as humorous or non-humorous). Of the examined transformer models, MuRIL has an advantage over the others because it was trained on both Indian languages and transliterated text, allowing it to better manage phonetic variation and code-mixing. The output from the transformer model is then passed to a classification layer to produce probability scores for each of the classes.

4.5 HEURISTIC ENGINE DESIGN

This engine uses a heuristic approach to identify potentially humorous non-standard linguistic features and other stylistic features associated with

humorous content. A few examples of the types of features tracked by the heuristic engine are: emojis, slang words/phrases, special tone markers (e.g., sarcasm), word patterns (e.g., repeated words, elongated words).

A score is assigned to all input to assess each feature. For instance, if input contains laughing emojis, or a slang word/phrase indicates that something may be humorous. The presence of sarcasm or words/phrases used to exaggerate or amplify an expression indicates a likelihood that something may be humorous and would therefore impact the heuristic score.

This engine serves as an adjunct to the Transformer model by providing additional information that may not be expressed via the contextual embeddings utilized by this model alone.

4.6 HYBRID PREDICTION MECHANISM

The final prediction is derived from both the Transformer and heuristic engine outputs and therefore a scoring mechanism exists for determining the contribution of each output.

If the heuristic score is above a defined threshold, it is used to override the prediction of the Transformer model. In circumstances where the heuristic signal is somewhat strong, both outputs are combined in a weighted manner. If not, the prediction is derived primarily from the output of the Transformer model.

This hybrid approach ensures that both the contextual understanding of the expression and heuristic cues are effectively utilized to increase accuracy and robustness of the results.

4.7 ALGORITHM WORKFLOW

The general workflow of the system can be explained in the following manner. The input text is first prepared (pre-processing) in order to break the text into token-sized units (tokenization). The tokenized text is then sent through the transformer model, to generate embeddings and initial predictions. While this is happening, the heuristic engine will also analyze the text to calculate a humour score based on predefined rules.

After both components have produced outputs, the outputs from both components are combined via a hybrid prediction mechanism to produce final classification results. As a result, this workflow allows the system to accurately handle linguistic and non-linguistic aspects of humour.

4.8 HANDLING CODE-MIXED TEXT

Due to the presence of multiple languages in a sentence when dealing with code-mixed text, there are some special techniques required to deal with these types of documents. Therefore, the system utilizes the ability of multilingual transformer models to analyze code-mixed input documents.

Furthermore, the preprocessing techniques of transliteration differences and text normalization give the heuristic engine more information about humour by identifying language-specific characteristics and patterns and expressions.

4.9 DESIGN CONSIDERATIONS

While designing the system, numerous design decision criteria were considered such as scalability, robustness, and adaptability for different types of data. Scalability would be determined through the use of trans-

former models while the heuristic engine allows for adaptation to nonstandard slang or newer evolving patterns of communication.

Another aspect for this design is its capacity to process real-world types of (often) noisy and unstructured data, therefore employing multiple approaches achieves the optimal balance between accuracy and generalization.

4.10 SUMMARY OF DESIGN AND METHODOLOGY

Finally, the combination of transformer models with an heuristic engine creates an environment conducive to successfully detecting humor within Tenglish text. In designing the system, attention has been given to solving the problems associated with code mixing language processing by utilizing both contextual and stylistic indicators. As the proposed hybrid approach improves performance and increases robustness, it will be able to achieve its intended goals.

Chapter 5

IMPLEMENTATION

5.1 OVERVIEW OF IMPLEMENTATION

The construction of the humor detection system described in the present work will be based on a hybrid approach utilizing transformer models combined with an engine based on heuristic techniques, in order to classify Romanized Telugu–English (Tenglish) text efficiently and accurately. Moreover, the system will be capable of processing informal, real-world, and code-mixed input with a high level of accuracy and reliability.

The implementation will consist of five components: preparation of Tenglish data, training transformer model(s), integrating the heuristic engine with the trained model(s), evaluating the model(s) against specified criteria, and finally, deploying the system to be used in production.

The humor detection system will be developed using modern machine learning libraries and frameworks, and the implementation will be designed for scalability and ease of experimentation. Furthermore, care has been taken to ensure that all transformer models have been designed with consistent settings, thus allowing for fair comparisons of system performance across different transformer architectures.

5.2 DEVELOPMENT ENVIRONMENT

Python was chosen as the primary programming language for the implementation of the humor detection system, as there are many libraries available for both machine learning and natural language processing in Python, making it easy to use. The PyTorch library will be used as the framework for training and fine-tuning transformer models, and the HuggingFace Transformers library will provide pre-trained transformer models and tokenizers.

In addition to these libraries, other common Python libraries will be used to handle and preprocess Tenglish data (e.g., NumPy and Pandas), and the Scikit-learn library will be used to measure evaluation metrics and compute performance statistics during the experimental evaluation of the humor detection system. The experiments will be run on a machine equipped with one or more NVIDIA GPUs to expedite both the training and inference processes.

5.3 DATASET PREPARATION

The implementation dataset for this study is made up of 10,000 Tenglish text samples spread evenly between the humorous and non-humorous categories. The dataset was preprocessed prior to use by eliminating all unwanted noise from the data, normalising the text of the data, and accounting for special character usage. Although emoji's and stylistic features were kept in the dataset, their retention was necessary in order to accurately detect humour.

The collected dataset was separated into three datasets; training, validation, and testing, proportioned as (80percent/ 10 percent/ 10 percent re-

spectively), using stratified random sampling, so as to be equally represented in each dataset. As part of a cross-dataset evaluation regarding the generalisation ability of the models, two versions of the dataset (V2 and V3) were used for evaluation purposes.

5.4 TEXT PREPROCESSING

In the process of cleaning up the data for use in model training the preprocessing of text is a very important step. The preprocessing pipeline not only requires the cleaning of the text but also the removal of all unnecessary characters and normalisation of the case of letters used. Due to the inclusion of code-mixed text within the dataset, it was also necessary to use a special methodology for maintaining the patterns of transliteration and the various types of linguistic diversity present within the text.

In order to accommodate the requirements articulated above each sequential (i.e., by sentence or paragraph) data token was generated using the appropriate tokenizer(s) associated with each transformer model. The utilisation of the respective tokenizer(s) allowed for the facilitation of the pre-trained word embeddings associated with all of the models to successfully process the data as intended. Finally, all tokens generated were either padded or truncated to ensure that all tokens had the same length (128 sequential tokens).

5.5 TRANSFORMER MODEL IMPLEMENTATION

The transformer-based model is the foundation of this system, and it is also the classifier. The various transformer models incorporated into this system include MuRIL, mBERT, XLM-RoBERTa, IndicBERT, IndicBERT v2,

Telugu-BERT, XLM, LaBSE, mT5 and ByT5. Each one of these transformer models has been trained via fine-tuning with the prepared dataset in binary classification.

The models are initialized with pretrained weights and will then be further trained on Tenglish. Each of the transformer models has had a classification head added on top of it to produce probability scores for the two classes. The optimization is accomplished by adapting model parameters utilizing back-propagation and minimization of cross-entropy loss.

Overall, MuRIL performs the best across the board because of its capability to handle Indian languages and transliterated text successfully. Each of these transformer models was trained under the same conditions in order to provide a fair comparison between all of the transformer models.

5.6 HEURISTIC ENGINE IMPLEMENTATION

Heuristic engine is a rule-based system that works by analyzing the input text to find specific characteristics associated with humor. The characteristics examined by the heuristic engine include (but are not limited to), emojis, slang, indications of sarcasm, and stylistic features such as elongated words or repeated letters. The engine uses these identified characteristics to assign a score to the input text, with higher scores indicating an increased likelihood of humor being present as a result of the quantity of characteristics exhibited in the input text. For example, an input with many laughing emoji will be associated with a higher score than an input that does not have any; likewise, an input exhibiting sarcasm or exaggeration will also receive a higher score.

The use of simple pattern matching and rule-based logic makes the implementation of this component straightforward to integrate with the

underlying transformer model.

5.7 HYBRID INTEGRATION

Integration between the heuristic engine and transformer model is achieved via a hybrid prediction methodology. The output probabilities from the transformer model are combined with the heuristic score to generate a final prediction.

A threshold-based method is used to determine the amount of influence the heuristic has on the prediction output. If the heuristic score is high, then the heuristic score will have a dominant influence over the output. If the heuristic score is moderate, then a weighted combination of both outputs will be used to generate a final output. If the heuristic score is low, then the final output will be based exclusively on the transformer model output.

Combining the predictions from the heuristic engine and the transformer model guarantees that the final prediction contains contextual features and stylistic features, resulting in improved accuracy for classifying humorous content.

5.8 TRAINING PROCESS

The fine-tuning of each transformer model utilizing the prepared dataset is considered to be the training cycle. Each model has been trained with a batch size and learning rate of 2×10^5 . The number of epochs is 3 to 5 depending upon when convergence occurs.

We use the Adam optimizer for the purpose of updating our model parameters, and after each epoch, we evaluate on a validation set and select the best-rated model based on the highest F1 score.

5.9 EVALUATION METRICS IMPLEMENTATION

Multiple metrics including accuracy, precision, recall and F1 score are all used to evaluate model performance and provide a complete evaluation.

Another metric used to evaluate the performance of the models with respect to their ability to handle code mixed text is by applying the degree of language mixing in the dataset as an index (Code-Mixing Index).

Confusion matrixes are also created in order to analyze the classification performance and to identify patterns of misclassification.

5.10 CROSS-DATASET EVALUATION IMPLEMENTATION

To evaluate generalization capability, cross dataset evaluation is accomplished by evaluating dataset versions V2 and V3. For each model trained on one dataset, we test the other dataset to evaluate performance with respect to new data distributions. This allows us to assess the robustness of the models' adaption to different language usage and style.

5.11 SYSTEM DEPLOYMENT

The project was developed using Gradio as the foundation for building an interactive interface to receive inputs from users in real-time and get predictions returned, along with some additional information including confidence scores and heuristic signals. The end product has an accessible and user-friendly interface making it suitable for many applications in the real world.

5.12 SUMMARY OF IMPLEMENTATION

The developed solution was able to present a successful implementation of a system combining transformer-based models with a heuristic engine to detect the humour in Tenglish-language text. As this approach is used to deal with the complexities associated with code-mixed languages, the hybridised implementation exhibits a high degree of performance. Using multiple models, implementing comprehensive evaluations, and providing a real-time system show the combined effectiveness and practicality of the resulting solution.

Chapter 6

HARDWARE AND SOFTWARE TOOLS USED

6.1 INTRODUCTION

For the proposed humor detection system to successfully incorporate the needed combination of hardware and software, an efficient way of processing, training, testing, and deploying data is needed. In order to ensure that the chosen hardware and software components will provide a system that is scalable, reliable, and performs as expected, specific sets of specifications or characteristics must be met. This chapter will provide a description of the hardware specifications and software technologies used to develop and execute the project. Several of the tools selected for use in developing this project are tools that are more widely used within machine learning and natural language processing; thus providing an excellent base on which to conduct experimentation and implementation.

6.2 HARDWARE REQUIREMENTS

The hardware configuration utilized for this project will support computationally intensive applications, such as training transformer-based models and processing large datasets. Therefore, the system has a multi-core processor, adequate memory and GPU support for accelerating the computation time of model execution and training.

For preprocessing and executing a model efficiently, it is recom-

mended that the system have at least an Intel Core i5 or equivalent processor (or higher) to handle these tasks efficiently; additionally, the availability of 8GB or more RAM will allow smooth operation when dealing with data-related tasks and model training processes. Training using deep learning methods (transformer models) on a system with GPU support is significantly faster than on a system without GPU support, thus reducing the time to train the model.

The experiments conducted in this project have utilized a system with GPU acceleration, which has resulted in accelerated fine-tuning of transformer models and optimized execution of multiple experiments. Therefore, the amount of storage available for your dataset, model checkpoint and output files is an important consideration as all these will require sufficient disk space.

6.3 SOFTWARE REQUIREMENTS

The software environment is a vital aspect to both creating machine learning models and controlling the entire workflow. The project is built using the Python programming language because it is so well known for supporting various data science and machine learning applications.

To develop and train deep learning models, we're using the PyTorch framework since it provides great flexibility as well as being easy to use. Also, we are using the HuggingFace Transformers library to access pre-trained transformer models such as MuRIL, mBERT, XLM-RoBERTa, among others; thus, making fine-tuning for the humor detection classification model easier and more efficient.

For data manipulation and preprocessing, we are using additional libraries, such as NumPy and Pandas. We will be using Scikit-learn to assist

us with evaluation metrics and performance evaluations. These libraries help provide us with essential functionalities for working with datasets and for evaluating the models.

6.4 DEVELOPMENT TOOLS

The project will be developed using integrated development environments (IDEs), which will help streamline the coding, debugging, and management processes of the projects. We will be using Visual Studio Code as our primary IDE because it provides a productive user interface for developers. Additionally, we will use Jupyter Notebook during experimentation and visualization; Jupyter Notebook allows us to interact with the model and perform analytical analyses of the data and model performance.

Git will be used for version control to help track changes and maintain integrity within the project; thus allowing us to maintain organization throughout our entire development processes.

6.5 LIBRARIES AND FRAMEWORKS

There are several custom libraries and libraries that can implement different system components. The HuggingFace Transformers are one of the most used libraries in this system. The PyTorch library is an important component and is used for developing and optimizing the machine learning model.

Gradio provides the means for building the user interface that allows users to view the predictions and results in real time by providing them with access to input text and output generated by the model. By allowing users to view prediction results in real time, users are able to engage more easily

with the system.

6.6 OPERATING SYSTEM

This implementation requires the use of a modern operating system that supports machine learning frameworks and GPU acceleration. This could be a Windows, Linux, or macOS based system, assuming the systems are all configured with the required dependencies and libraries. The operating system will be configured to run Python and CUDA (for GPU acceleration), as well as a variety of machine learning libraries. Proper configuration of the operating system will allow for efficient execution of the application and full compatibility with all of the tools that use the tools in this project.

6.7 SUMMARY OF TOOLS USED

The development and evaluation of this humor detection system are based on a set of hardware and software tools that can be used in combination to provide a complete environment for these activities. The hardware component consisted of a GPU-based hardware implementation, which provided fast training of the model. In addition, the software component provided advanced software frameworks for efficient implementation and scalability. All of these tools work together to make possible the successful execution of the project and to produce very high-quality results.

Chapter 7

RESULTS & DISCUSSION

7.1 INTRODUCTION

This chapter discusses the experimental results derived from the humor detection system proposed, and gives an in depth description regarding the findings. Multiple transformer-based approaches were assessed through their performance using standard metrics of evaluation (e.g: performance, precision, recall, and F1 score). Additionally, assessment of the effectiveness of transformer based models were compared against the Code-Mixing Index (CMI); more specifically, reference to the Code-Mixing Index (CMI) was used to evaluate the effectiveness of transformer based models for code-mixed content. A detailed analysis of the performance of a hybrid system meets multiple transformer-based learning methodologies and heuristic engines is provided. The findings establish the hybrid approaches effectiveness for improved humor detection performance with Romanized Telugu-English text.

7.2 TRANSFORMER MODEL COMPARISON

Ten transformer-based models were evaluated for effective similarities of training conditions to allow for a fair comparison. The transformer-based approaches include MuRIL, mBERT, XLM-RoBERTa, IndicBERT, IndicBERT v2, Telugu-BERT models, XLM, LaBSE, mT5, and ByT5.. All models un-

derwent the fine-tuning process applied to the entire data set; afterwards, tested on their respective testing sets.

Table 7.1: Performance Comparison of Transformer Models

Model	Accuracy (%)	Precision	Recall	F1 Score
MuRIL + Heuristic	92.4	0.912	0.925	0.918
MuRIL	90.1	0.895	0.902	0.898
XLM-RoBERTa	88.5	0.874	0.888	0.881
IndicBERT v2	87.9	0.868	0.876	0.872
IndicBERT	87.2	0.865	0.871	0.868
mBERT	85.8	0.842	0.860	0.851
Telugu-BERT	84.6	0.831	0.845	0.838
XLM	83.2	0.820	0.832	0.826
LaBSE	82.4	0.812	0.825	0.818
mT5	81.6	0.805	0.818	0.811
ByT5	80.9	0.798	0.810	0.804

Based on the findings of these experiments, we have determined that of all the transformer based models that we examined In our experimentation, the performance of the MuRIL based model was determined to be higher than any other transformer based model evaluated in this study. In addition, the fusion of the Heuristic-based Engine with the Transformer-based Model was shown to improve the Accuracy and F1 Score of the models considered. This indicates that the Heuristic-based Engine effectively improves the humor detection performance of the Transformer based models.

7.3 GRAPHICAL ANALYSIS

Graphical representations have been used to visualize the comparative performance of various models. Accuracy, which is a performance metric of the models, can be found in Figure 7.1; while a plot of how the F1-score measured for each model varied can be located in Figure 7.2.

Precision and recall can be compared and contrasted via their plotted values as displayed in Figure 7.3; this comparison will also illustrate how

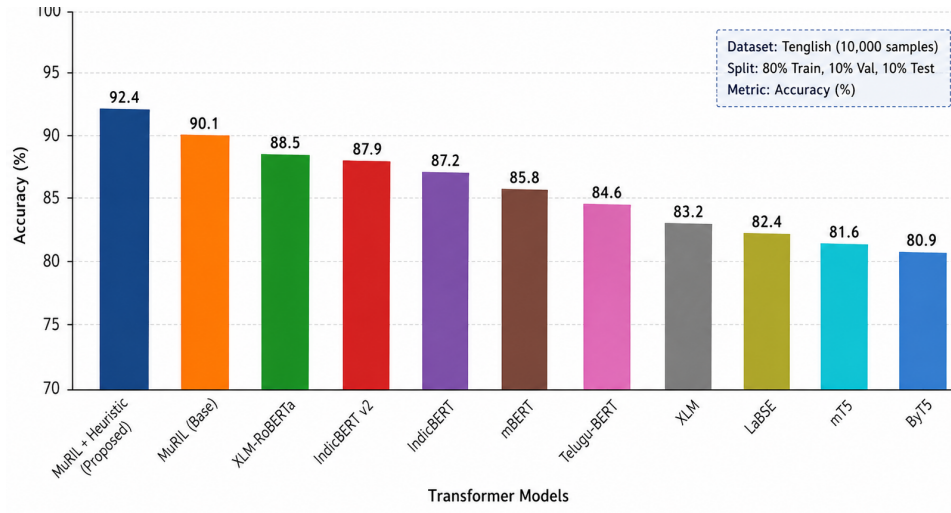


Figure 7.1: Accuracy comparison across transformer models

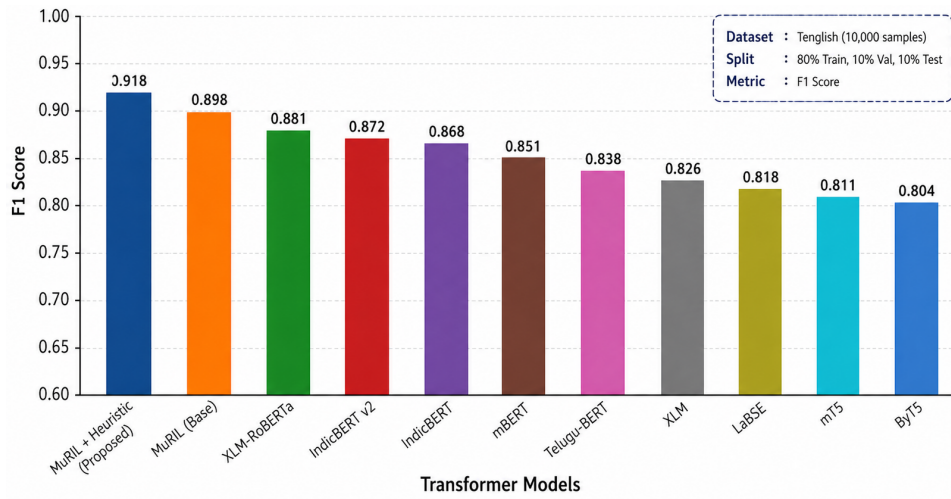


Figure 7.2: F1 score comparison across transformer models

balanced precision and recall are for each model being evaluated. All models being evaluated have been assessed for cross-dataset evaluation, which can be seen in Figure 7.4.

7.4 CROSS-DATASET EVALUATION

The study evaluated the models' generalization capabilities through cross-dataset experiments using V2 and V3 datasets. The models trained on

Model	MuRIL + Heuristic (Proposed)	MuRIL (Base)	XLM- RoBERTa	IndicBERT v2	IndicBERT	mBERT	Telugu- BERT	XLM	LaBSE	mT5	ByT5
Precision	0.912	0.895	0.874	0.868	0.865	0.842	0.831	0.820	0.812	0.805	0.798
Recall	0.925	0.902	0.888	0.876	0.871	0.860	0.845	0.832	0.825	0.818	0.810

Figure 7.3: Precision vs Recall comparison

Model	Train V2 → Test V3 (Accuracy %)		Train V3 → Test V2 (Accuracy %)		Average (Both Directions)	
	Accuracy	F1 Score	Accuracy	F1 Score	Accuracy	F1 Score
MuRIL + Heuristic (Proposed)	89.8	0.891	90.2	0.895	90.0	0.893
MuRIL (Base)	87.4	0.874	87.9	0.880	87.7	0.877
XLM-RoBERTa	86.1	0.861	86.7	0.867	86.4	0.864
IndicBERT v2	85.0	0.849	85.6	0.855	85.3	0.852
IndicBERT	84.2	0.840	84.8	0.846	84.5	0.843
mBERT	82.6	0.824	83.1	0.830	82.9	0.827
Telugu-BERT	81.3	0.813	81.9	0.819	81.6	0.816
XLM	80.2	0.802	80.8	0.807	80.5	0.805
LaBSE	79.1	0.791	79.8	0.797	79.5	0.794
mT5	77.8	0.778	78.5	0.784	78.2	0.781
ByT5	76.6	0.764	77.2	0.771	76.9	0.768

Figure 7.4: Cross-dataset evaluation results

one dataset were tested with the other dataset to determine how effectively they would generalize to the newly unseen distributions of data.

Table 7.2: Cross-Dataset Evaluation Results		
Training Dataset	Testing Dataset	Accuracy (%)
V2	V3	89.8
V3	V2	90.2

The results reflect a slight decline in model performance, as compared to evaluations on the same dataset; however, the overall performance of all models remained consistently high. This demonstrates the robustness of the hybrid model developed in this study.

7.5 CONFUSION MATRIX ANALYSIS

The confusion matrix allows for a more thorough evaluation of classification performance by illustrating the distribution of true and false predictions (both true and false positives).

From the confusion matrix, it can be concluded that the hybrid approach is effective in discriminating between humorous and non-humorous text due to the large proportion of accurate predictions (both true positives and true negatives), along with small numbers of erroneous predictions.

Table 7.3: Confusion Matrix (MuRIL + Heuristic)

	Predicted Humor	Predicted Non-Humor
Actual Humor	462	38
Actual Non-Humor	42	458

7.6 DISCUSSION OF RESULTS

Humor Detection in Code-Mixed Texts using Transformer-based Models: A Comparative Study; Findings The results of our experiments support the conclusion that transformer-based models can be effectively utilized to detect humour in code-mixed text. Among all the models tested, MuRIL emerges as the optimal solution due to its prior experience with training on Indian language resources and transliterated datasets. Moreover, incorporation of a heuristic engine provided a considerable boost to performance by extracting non-linguistic characteristics (e.g., emojis, slang) of the data. Such attributes were found to be essential for accurately identifying humorous content and thus provide context to the transformer-based models’ abilities to understand the intent behind a given piece of code-mixed text. Multilingual models (e.g., XLM-RoBERTa, IndicBERT) demonstrated robust performance levels; however, their ability to process variations resulting from transliteration was weaker than that exhibited by MuRIL model. Furthermore, the sequence-to-sequence architecture models (mT5, ByT5) performed worse than both MuRIL and the multilingual architectures. Therefore, we recommend against using these architectures for classification tasks in the current domain. Finally, the cross-dataset evaluation results provided validation of the generalizability of our proposed model across multiple data distributions; this facet is critical when considering real world applications where there exists significant variability in the underlying content.

7.7 SUMMARY

The results from the proposed humour detection system were presented in detail in this chapter and indicated that the hybrid approach was very effective. The MuRIL based model was also demonstrated to be superior to other methods. Results highlight the need for handling code-mixed language, and utilising heuristic features.

Chapter 8

CONCLUSION

This study investigated an issue concerning humor detection in Romanized Telugu-English code-mixed text (Tenglish) due to the significant increase in usage of social networking sites and informal digital communication, code-mixed language has developed as a major vehicle for communication in Asia and particularly in India, a multilingual country. Here traditional natural language processing (NLP) systems face difficulties when processing such texts as they typically do not deal well with the different complexities present in Tenglish: varying transliterations, informal grammar, and the multiple languages within one sentence.

To accommodate these complexities a hybrid method combining transformer models with a rule based heuristics engine was developed. The transformers captured the contextual and semantic features of the text, which were then complemented by the heuristics identifying additional humor specific signals found throughout the text (such as emojis/slang terms/sarcasm patterns/stylistic differences, etc.). Through the use of this combined technique, the humor detection system could leverage both the deep learning power of the transformer models and domain-specific linguistic knowledge.

A thorough evaluation was done with a balanced dataset of 10,000 samples, made up of both humorous and non-humorous Tenglish text. We used several transformer models, including MuRIL, mBERT, XLM-RoBERTa, IndicBERT, IndicBERT v2, Telugu-BERT, XLM, LaBSE, mT5, and

ByT5, and compared their performances under the same experimental conditions. Out of these, the MuRIL model performed the best because it was trained on Indian languages and transliterated text, making it especially good at handling Tenglish inputs.

Adding the heuristic engine improved the system's performance even more, achieving a maximum accuracy of 92.4 percent. The results clearly show that including non-linguistic and stylistic features greatly boosts humor detection. Additionally, cross-dataset evaluation using dataset versions V2 and V3 proved the strength and generalization ability of the proposed system. Even with differences in data distribution, the model kept high accuracy, showing it is suitable for real-world use.

The findings of this project show the importance of creating specialized methods for processing code-mixed languages. While transformer models offer strong contextual understanding, they might not capture all aspects of informal communication. Adding heuristic techniques helps close this gap and leads to better performance.

Overall, the proposed hybrid humor detection system offers an effective way to analyze Tenglish text. It tackles significant challenges in code-mixed NLP and shows the potential of merging machine learning models with rule-based improvements. The system can be expanded to other code-mixed languages and used in various applications like sentiment analysis, chatbot development, and social media monitoring.

In conclusion, this work adds to the growing field of multilingual NLP by presenting a solid and scalable method for humor detection in code-mixed text. The results confirm the effectiveness of the proposed approach and provide a strong basis for future research in this area.

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